

Research Monographaskdfjhjk

Book Author

Month Year

Mục lục

Preface

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part I.

First half of the book

chapter 1.

Introduction

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1.1. Research question

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1.2. Answer to the question

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1.3. Positioning in the literature

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1.4. A section

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1.4.2. Yet another subsection

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chapter 2.

Sections and content

part II.

Thiết kế logic

chapter 3.

thuật toán 2.2 mapping ER

part III.

Thiet Ke Vat Ly

chapter 4.

Thiết Kế Vật LÝ

chapter 5.

Thiết kế vật lý

5.1. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

5.1.1. Cú pháp Rút gọn (Logical Schema Syntax)

- templates (template_id UUID PK, workspace_id UUID FK, name VARCHAR(150), description TEXT, content_structure TEXT, creator_type VARCHAR(20), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP)
- template_sections (section_id UUID PK, template_id UUID FK, heading_title VARCHAR(255), order_index INTEGER, prompt_instruction TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- placeholders (placeholder_id UUID PK, section_id UUID FK, variable_name VARCHAR(100), expected_data_type VARCHAR(30), description TEXT, is_required BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- template_mappings (mapping_id UUID PK, placeholder_id UUID FK, code_entity_id UUID FK, extraction_rule TEXT, is_active BOOLEAN)
- documents (document_id UUID PK, workspace_id UUID FK, author_user_id UUID FK, title VARCHAR(255), document_type VARCHAR(30), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)

- document_versions (version_id UUID PK, document_id UUID FK, version_number VARCHAR(20), content TEXT, change_summary TEXT, status VARCHAR(20), ticket_codes VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)
- change_logs (log_id UUID PK, version_id UUID FK, change_type VARCHAR(20), change_description TEXT, diff_content TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)

5.1.2. Mã DDL SQL Hoàn Chỉnh (schema_member_4.sql)

```

1
2
3
4 CREATE TABLE templates (
5     template_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
6     workspace_id         UUID          NOT NULL,
7     name                 VARCHAR(150)  NOT NULL,
8     description          TEXT          NULL,
9     content_structure     TEXT          NULL,
10    creator_type         VARCHAR(20)   NOT NULL,
11    status               VARCHAR(20)   NOT NULL DEFAULT 'DRAFT' ,
12    created_at           TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
13    updated_at           TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
14    CONSTRAINT pk_templates PRIMARY KEY (template_id) ,
15
16    CONSTRAINT uq_templates_workspace_name UNIQUE (workspace_id, name) ,
17    CONSTRAINT fk_templates_workspace
18        FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
19        ON UPDATE CASCADE
20        ON DELETE CASCADE,
21    CONSTRAINT ck_templates_creator_type
22        CHECK (creator_type IN ( 'SYSTEM' , 'USER' , 'AI' )),
23    CONSTRAINT ck_templates_status
24        CHECK (status IN ( 'DRAFT' , 'ACTIVE' , 'ARCHIVED' )),
25    CONSTRAINT ck_templates_name_not_blank
26        CHECK (length(btrim(name)) > 0) ,
27    CONSTRAINT ck_templates_updated_after_created
28        CHECK (updated_at >= created_at)
29 );
30
31 CREATE TABLE template_sections (
32     section_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
33     template_id         UUID          NOT NULL,
34     heading_title       VARCHAR(255)  NOT NULL,
35     order_index         INTEGER       NOT NULL,
36     prompt_instruction  TEXT          NULL,

```

```

37     created_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
38     updated_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
39     CONSTRAINT pk_template_sections PRIMARY KEY (section_id),
40
41     CONSTRAINT uq_template_sections_order UNIQUE (template_id, order_index),
42     CONSTRAINT fk_template_sections_template
43         FOREIGN KEY (template_id) REFERENCES templates (template_id)
44         ON UPDATE CASCADE
45         ON DELETE CASCADE,
46     CONSTRAINT ck_template_sections_order_index
47         CHECK (order_index >= 0),
48     CONSTRAINT ck_template_sections_heading_not_blank
49         CHECK (length(btrim(heading_title)) > 0)
50 );
51
52 CREATE TABLE placeholders (
53     placeholder_id      UUID                NOT NULL DEFAULT gen_random_uuid(),
54     section_id          UUID                NOT NULL,
55     variable_name       VARCHAR(100)        NOT NULL,
56     expected_data_type  VARCHAR(30)         NOT NULL,
57     description         TEXT                NULL,
58     is_required         BOOLEAN             NOT NULL DEFAULT TRUE,
59     created_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
60     updated_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
61     CONSTRAINT pk_placeholders PRIMARY KEY (placeholder_id),
62
63     CONSTRAINT uq_placeholders_section_variable UNIQUE (section_id,
64     variable_name),
65     CONSTRAINT fk_placeholders_section
66         FOREIGN KEY (section_id) REFERENCES template_sections (section_id)
67         ON UPDATE CASCADE
68         ON DELETE CASCADE,
69     CONSTRAINT ck_placeholders_expected_data_type
70         CHECK (expected_data_type IN ('STRING', 'NUMBER', 'BOOLEAN',
71         'DATE', 'LIST', 'OBJECT', 'CODE')),
72
73     CONSTRAINT ck_placeholders_variable_name_format
74         CHECK (variable_name ~ '^[A-Za-z_][A-Za-z0-9_]*$')
75 );
76
77 CREATE TABLE template_mappings (
78     mapping_id          UUID                NOT NULL DEFAULT gen_random_uuid(),
79     placeholder_id      UUID                NOT NULL,
80     code_entity_id      UUID                NOT NULL,
81     extraction_rule     TEXT                NULL,
82     is_active           BOOLEAN             NOT NULL DEFAULT TRUE,
83     CONSTRAINT pk_template_mappings PRIMARY KEY (mapping_id),

```

```

83
84     CONSTRAINT uq_template_mappings_pair UNIQUE (placeholder_id ,
85     code_entity_id) ,
86     CONSTRAINT fk_template_mappings_placeholder
87         FOREIGN KEY (placeholder_id) REFERENCES placeholders
88         (placeholder_id)
89         ON UPDATE CASCADE
90         ON DELETE CASCADE,
91     CONSTRAINT fk_template_mappings_code_entity
92         FOREIGN KEY (code_entity_id) REFERENCES code_entities
93         (code_entity_id)
94         ON UPDATE CASCADE
95         ON DELETE CASCADE
96 );
97
98 CREATE TABLE documents (
99     document_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
100     workspace_id         UUID          NOT NULL,
101     author_user_id       UUID          NOT NULL,
102     title                VARCHAR(255)  NOT NULL,
103     document_type        VARCHAR(30)   NOT NULL,
104     status               VARCHAR(20)   NOT NULL DEFAULT 'DRAFT' ,
105     created_at           TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
106     updated_at           TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
107     deleted_at           TIMESTAMP      NULL,
108     CONSTRAINT pk_documents PRIMARY KEY (document_id) ,
109     CONSTRAINT fk_documents_workspace
110         FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
111         ON UPDATE CASCADE
112         ON DELETE CASCADE,
113     CONSTRAINT fk_documents_author
114         FOREIGN KEY (author_user_id) REFERENCES users (user_id)
115         ON UPDATE CASCADE
116         ON DELETE RESTRICT,
117     CONSTRAINT ck_documents_type
118         CHECK (document_type IN ( 'SRS' , 'API_DOC' , 'USER_GUIDE' ,
119         'RELEASE_NOTE' , 'TECH_DESIGN' , 'OTHER' )),
120     CONSTRAINT ck_documents_status
121         CHECK (status IN ( 'DRAFT' , 'IN_REVIEW' , 'PUBLISHED' , 'ARCHIVED' )),
122     CONSTRAINT ck_documents_title_not_blank
123         CHECK (length(btrim(title)) > 0) ,
124     CONSTRAINT ck_documents_deleted_after_created
125         CHECK (deleted_at IS NULL OR deleted_at >= created_at)
126 );
127
128 CREATE UNIQUE INDEX uq_documents_workspace_title_active
129     ON documents (workspace_id, title)

```

```

127     WHERE deleted_at IS NULL;
128
129 CREATE TABLE document_versions (
130     version_id          UUID          NOT NULL DEFAULT gen_random_uuid() ,
131     document_id         UUID          NOT NULL,
132     version_number      VARCHAR(20)   NOT NULL,
133     content             TEXT          NULL,
134     change_summary      TEXT          NULL,
135     status              VARCHAR(20)   NOT NULL DEFAULT 'DRAFT' ,
136     ticket_codes        VARCHAR(500)  NULL,
137     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
138     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
139     CONSTRAINT pk_document_versions PRIMARY KEY (version_id) ,
140
141     CONSTRAINT uq_document_versions_number UNIQUE (document_id ,
142     version_number) ,
143     CONSTRAINT fk_document_versions_document
144         FOREIGN KEY (document_id) REFERENCES documents (document_id)
145         ON UPDATE CASCADE
146         ON DELETE CASCADE,
147     CONSTRAINT ck_document_versions_status
148         CHECK (status IN ( 'DRAFT' , 'IN_REVIEW' , 'APPROVED' , 'REJECTED' ,
149         'PUBLISHED' ) ) ,
150
151     CONSTRAINT ck_document_versions_number_format
152         CHECK (version_number ~ '^[vV]?[0-9]+(\.[0-9]+){0,2}$' )
153 );
154
155 CREATE TABLE change_logs (
156     log_id              UUID          NOT NULL DEFAULT gen_random_uuid() ,
157     version_id          UUID          NOT NULL,
158     change_type         VARCHAR(20)   NOT NULL,
159     change_description  TEXT          NULL,
160     diff_content        TEXT          NULL,
161     created_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
162     updated_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
163     CONSTRAINT pk_change_logs PRIMARY KEY (log_id) ,
164     CONSTRAINT fk_change_logs_version
165         FOREIGN KEY (version_id) REFERENCES document_versions (version_id)
166         ON UPDATE CASCADE
167         ON DELETE CASCADE,
168     CONSTRAINT ck_change_logs_change_type
169         CHECK (change_type IN ( 'CREATE' , 'UPDATE' , 'DELETE' , 'RESTORE' ,
170         'PUBLISH' ) )
171 );

```

```

171 CREATE INDEX idx_templates_workspace_id ON templates (workspace_id);
172 CREATE INDEX idx_templates_status ON templates (status);
173
174 CREATE INDEX idx_template_mappings_code_entity ON template_mappings
    (code_entity_id);
175 CREATE INDEX idx_template_mappings_active ON template_mappings
    (placeholder_id)
176 WHERE is_active = TRUE;
177
178 CREATE INDEX idx_documents_workspace_id ON documents (workspace_id);
179 CREATE INDEX idx_documents_author ON documents (author_user_id);
180 CREATE INDEX idx_documents_status ON documents (status);
181 CREATE INDEX idx_documents_created_at ON documents (created_at DESC);
182
183 CREATE INDEX idx_document_versions_status ON document_versions (status);
184 CREATE INDEX idx_document_versions_created_at ON document_versions
    (document_id, created_at DESC);
185
186 CREATE INDEX idx_change_logs_version_id ON change_logs (version_id);
187 CREATE INDEX idx_change_logs_created_at ON change_logs (created_at DESC);
188
189 CREATE OR REPLACE FUNCTION set_updated_at()
190 RETURNS TRIGGER AS $$
191 BEGIN
192     NEW.updated_at := CURRENT_TIMESTAMP;
193     RETURN NEW;
194 END;
195
196 CREATE TRIGGER trg_templates_updated_at
197 BEFORE UPDATE ON templates
198 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
199 CREATE TRIGGER trg_template_sections_updated_at
200 BEFORE UPDATE ON template_sections
201 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
202 CREATE TRIGGER trg_placeholders_updated_at
203 BEFORE UPDATE ON placeholders
204 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
205 CREATE TRIGGER trg_documents_updated_at
206 BEFORE UPDATE ON documents
207 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
208 CREATE TRIGGER trg_document_versions_updated_at
209 BEFORE UPDATE ON document_versions
210 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
211 CREATE TRIGGER trg_change_logs_updated_at
212 BEFORE UPDATE ON change_logs
213 FOR EACH ROW EXECUTE FUNCTION set_updated_at();
214 GO

```

5.2. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

5.2.1. Cú pháp Rút gọn (Logical Schema Syntax)

- reviews (review_id UUID PK, version_id UUID FK, user_id UUID FK, status VARCHAR(20), decision VARCHAR(20), summary_notes TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- review_comments (comment_id UUID PK, review_id UUID FK, user_id UUID FK, content TEXT, is_resolved BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- approvals (approval_id UUID PK, version_id UUID FK, user_id UUID FK, decision VARCHAR(20), rejection_reason TEXT, approved_at TIMESTAMP)
- grounding_evidences (evidence_id UUID PK, version_id UUID FK, snippet_content TEXT, purpose VARCHAR(30), created_at TIMESTAMP)
- drift_alerts (alert_id UUID PK, evidence_id UUID FK, repository_id UUID FK, drift_type VARCHAR(20), status VARCHAR(20), detected_at TIMESTAMP)
- reports (report_id UUID PK, workspace_id UUID FK, report_type VARCHAR(50), metric_data TEXT, generated_at TIMESTAMP)

5.2.2. Mã DDL SQL Hoàn Chỉnh (schema_member_5.sql)

```
1
2
3 CREATE TABLE reviews (
4     review_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
5     version_id UNIQUEIDENTIFIER NOT NULL,
6     user_id UNIQUEIDENTIFIER NOT NULL,
7     status VARCHAR(20) NOT NULL DEFAULT 'PENDING' ,
8     decision VARCHAR(20) NULL,
9     summary_notes NVARCHAR(MAX) NULL,
10    created_at DATETIME2 NOT NULL DEFAULT GETDATE() ,
11    updated_at DATETIME2 NULL,
12
13    CONSTRAINT chk_reviews_status CHECK (status IN ( 'PENDING' , 'IN_PROGRESS'
14    , 'COMPLETED' ) ) ,
15    CONSTRAINT chk_reviews_decision CHECK (decision IS NULL OR decision IN (
16    'APPROVED' , 'REJECTED' , 'CHANGES_REQUESTED' ) )
17 );
```

```

17 CREATE TABLE review_comments (
18     comment_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
19     review_id UNIQUEIDENTIFIER NOT NULL,
20     user_id UNIQUEIDENTIFIER NOT NULL,
21     content NVARCHAR(MAX) NOT NULL,
22     is_resolved BIT NOT NULL DEFAULT 0,
23     created_at DATETIME2 NOT NULL DEFAULT GETDATE() ,
24     updated_at DATETIME2 NULL,
25
26     CONSTRAINT fk_comments_review FOREIGN KEY (review_id)
27         REFERENCES reviews(review_id) ON DELETE CASCADE
28 );
29
30 CREATE TABLE approvals (
31     approval_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
32     version_id UNIQUEIDENTIFIER NOT NULL,
33     user_id UNIQUEIDENTIFIER NOT NULL,
34     decision VARCHAR(20) NOT NULL,
35     rejection_reason NVARCHAR(MAX) NULL,
36     approved_at DATETIME2 NOT NULL DEFAULT GETDATE() ,
37
38     CONSTRAINT chk_approvals_decision CHECK (decision IN ( 'APPROVED', '
39     REJECTED'))
40 );
41
42 CREATE TABLE grounding_evidences (
43     evidence_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
44     version_id UNIQUEIDENTIFIER NOT NULL,
45     snippet_content NVARCHAR(MAX) NOT NULL,
46     purpose VARCHAR(30) NOT NULL DEFAULT 'HALLUCINATION_PREVENTION' ,
47     created_at DATETIME2 NOT NULL DEFAULT GETDATE() ,
48
49     CONSTRAINT chk_evidences_purpose CHECK (purpose IN ( '
50     HALLUCINATION_PREVENTION', 'CODE_REFERENCE', 'AST_VERIFICATION'))
51 );
52
53 CREATE TABLE drift_alerts (
54     alert_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
55     evidence_id UNIQUEIDENTIFIER NOT NULL,
56     repository_id UNIQUEIDENTIFIER NOT NULL,
57     drift_type VARCHAR(20) NOT NULL,
58     status VARCHAR(20) NOT NULL DEFAULT 'OPEN' ,
59     detected_at DATETIME2 NOT NULL DEFAULT GETDATE() ,
60
61     CONSTRAINT fk_alerts_evidence FOREIGN KEY (evidence_id)
62         REFERENCES grounding_evidences(evidence_id) ON DELETE CASCADE,
63     CONSTRAINT chk_alerts_type CHECK (drift_type IN ( 'REFERENTIAL', '

```

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SIGNATURE', 'SEMANTIC', 'OUTDATED'))),
62     CONSTRAINT chk_alerts_status CHECK (status IN ('OPEN', 'RESOLVED', '
        IGNORED'))
63 );
64
65 CREATE TABLE reports (
66     report_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID() ,
67     workspace_id UNIQUEIDENTIFIER NOT NULL,
68     report_type VARCHAR(50) NOT NULL,
69     metric_data NVARCHAR(MAX) NOT NULL,
70     generated_at DATETIME2 NOT NULL DEFAULT GETDATE()
71 );
72 GO
73
74 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_version
75     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
76 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_user
77     FOREIGN KEY (user_id) REFERENCES users(user_id);
78 ALTER TABLE review_comments ADD CONSTRAINT fk_comments_user
79     FOREIGN KEY (user_id) REFERENCES users(user_id);
80 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_version
81     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
82 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_user
83     FOREIGN KEY (user_id) REFERENCES users(user_id);
84 ALTER TABLE grounding_evidences ADD CONSTRAINT fk_evidences_version
85     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
86 ALTER TABLE drift_alerts ADD CONSTRAINT fk_alerts_repository
87     FOREIGN KEY (repository_id) REFERENCES repositories(repository_id);
88 ALTER TABLE reports ADD CONSTRAINT fk_reports_workspace
89     FOREIGN KEY (workspace_id) REFERENCES workspaces(workspace_id);
90 GO

```


chapter 6.

Extensions

part IV.

Conclusion

chapter 7.

Summary

chapter 8.

Future research

part V.

Technical appendix

appendix A.

Generic technical topic

appendix B.

Another technical topic

appendix C.

Last technical topic

This is a last appendix with a bit of text.

C.1. A first section

Sed vestibulum ex a tristique lacinia. Integer interdum magna vel magna rutrum fermentum. Maecenas sed mi in nunc convallis rutrum. Vivamus dapibus bibendum est, ac tincidunt ipsum fermentum id. Nunc id est turpis. Suspendisse potenti. Sed ac laoreet nulla, eu sollicitudin libero. Suspendisse tempus orci nec mauris volutpat eleifend. Integer nec tristique libero. Sed vehicula ipsum sit amet magna accumsan, nec pulvinar nisl consequat. In ac hendrerit turpis. Sed aliquet luctus mauris, vitae congue turpis. Nullam blandit lacus vel interdum eleifend.

C.2. A second section

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$(C.1) \quad S^*(z) = \frac{\alpha}{1 - \gamma(1 - \alpha)} > \int \frac{\alpha(i)}{\gamma(i)} di.$$

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$(C.2) \quad \mathcal{W}^2(\zeta) = \left[\frac{\beta}{1 - \nu(1 - \xi)} \right] \pm \Lambda$$

$$(C.3) \quad \mathcal{L}^2(\zeta) = \left(\frac{\beta}{1 - \nu(1 - \xi)} \right)$$

$$(C.4) \quad \overline{a - b} = \int_0^{+\infty} \frac{\beta}{1 - \nu(x)(1 - \xi)} dx \in \mathbb{R}.$$

C.3. A third section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed aliquam magna vel urna ultrices, sed lacinia nulla mattis. Fusce non nunc nec est mollis malesuada. Nullam dignissim nulla sit amet libero facilisis, eget fringilla libero sagittis. Suspendisse potenti. Vivamus fermentum consectetur ante, at rhoncus nisi tristique vel. Vivamus in est quis justo fermentum lacinia ac eu leo. Sed ac pulvinar nulla. Etiam quis felis dapibus, vulputate metus eu, finibus nunc. Sed vel sodales dui. Nam venenatis dolor non orci tempus fermentum. Vivamus sodales justo a ligula cursus aliquet. Sed fringilla nunc vitae justo finibus, id placerat lectus sodales.

Tài liệu tham khảo

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Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum, quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. At vero eos et accusamus et iusto odio dignissimos ducimus, qui blanditiis praesentium voluptatum deleniti atque corrupti, quos dolores et quas molestias excepturi sint, obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga.

Future research